



EAST WEST UNIVERSITY

Project on
**Database Design & SQL implementation: Online Shopping
(E-commerce) System**

Course Code: DSA 201
Course Title: Data Processing and Storage
Spring 2026

Submitted To:

Sudarshan Saha

Lecturer
Department of Mathematical and Physical Sciences
East West University

Submitted by:

Tasmia Rahman Autoshi

ID: 2024-3-27-066
Section: 02

Submission Deadline: 20-04-2026

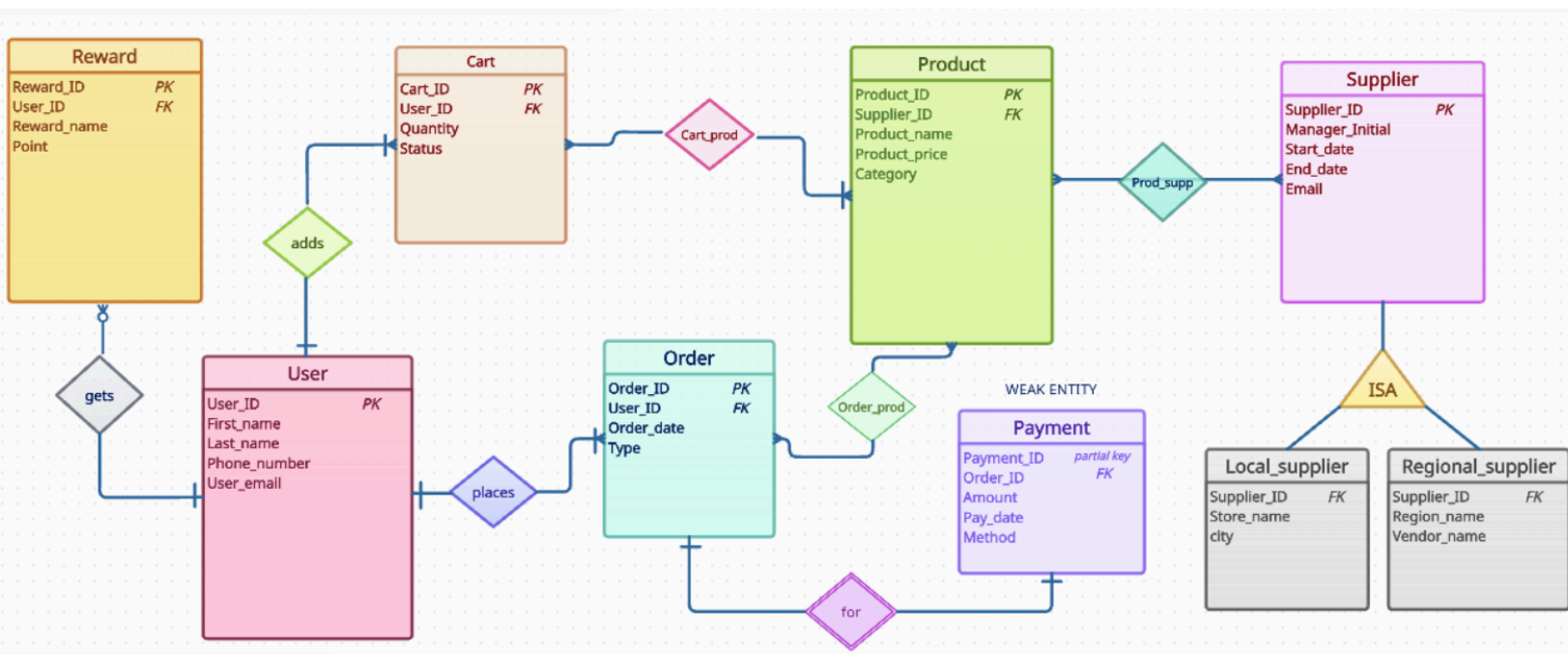
Table of contents

ER Model: B2C Online Grocery Shopping System.....	2
System Description & Entities, Attributes, and Relationships:.....	3
Relational Schema:.....	3
Specialization.....	4
Normalization:.....	4
1NF Rule:.....	4
2NF Rule:.....	5
3NF Rule:.....	5
BCNF Rule:.....	5
4NF Rule:.....	5
SQL implementation.....	6
1. Database Creation and Select:.....	6
2. `User` Table Creation:.....	6
3. Insertion Records in User table:.....	7
4. Creation & Insertion of `Reward` Table:.....	7
5. Creation & Insertion of `Cart` Table:.....	8
6. Creation & Insertion of `Order` Table:.....	9
7. Creation & Insertion of `Supplier` Table:.....	10
8. Creation & Insertion of `Local_supplier` Table:.....	11
9. Creation & Insertion of `Regional_supplier` Table:.....	12
10. Creation & Insertion of `Product` Table:.....	13
11. Creation & Insertion of `Payment` Table:.....	13
12. Creation & Insertion of `Cart_prod` Table:.....	14
13. Creation & Insertion of `Orer_prod` Table:.....	15
14. Creation & Insertion of `Prod_supp` Table:.....	16
15. `egrocery_shopping_bd` Database View:.....	17
Some Query from `egrocery_shopping_bd` Database:.....	18
Using LEFT JOIN.....	18
Using INNER JOIN.....	19
Using Inner Join with Conditions.....	20
Simple Query Example.....	21
Using GROUP BY.....	21
Using Aggregated Function.....	22
Using Sub-Query.....	24

DSA201: Data Processing & Storage

Database Design Project Assignment

ER Model: B2C Online Grocery Shopping System



System Description & Entities, Attributes, and Relationships:

This ER model represents a B2C (Business to Customer) online grocery shopping system where users purchase many grocery products provided by suppliers. It has entities such as **User**, **Reward**, **Cart**, **Order**, **Product**, **Payment** and **Supplier** (sub-entities **Local** and **Regional**) with relevant attributes and primary keys.

- Each User can have many Carts and can place multiple Orders
- Each Order belongs to one user and has exactly one Payment (weak entity)
- Each Order includes multiple Products and each product can appear in multiple orders.
(M:N)
- A supplier can supply many products and a product can be supplied by many suppliers.
(M:N)
- One cart can contain many products and one product can appear in many different carts.
(M:N)
- Supplier is specialized into Local_supplier and Regional_supplier. (ISA hierarchy)

Relational Schema:

- User (**User_ID**, First_name, Last_name, Phone_number, User_email)
- Reward (**Reward_ID**, User_ID (FK), Reward_name, Point)
- Cart (**Cart_ID**, User_ID (FK), Quantity, Status)
- Cart_prod (**Cart_ID**, **Product_ID**)
[Cart_ID FK from Cart, Product_ID FK from Product,
(Cart_ID, Product_ID) Component key for 'Cart_prod' entity]
- Order (**Order_ID**, User_ID (FK), Order_date, Type)
- Order_prod (**Order_ID**, **Product_ID**)

[Order_ID FK from Order , Product_ID FK from Product

(Order_ID, Product_ID) Component key for 'Order_prod' entity]

- Payment (**Payment_ID** (discriminator), **Order_ID** (FK), Amount, Pay_date, Method)
- Product (**Product_ID**, Supplier_ID (FK), Product_name, Price, Category)
- Prod_supp (**Product_ID, Supplier_ID**)

[Product_ID FK from Product , Supplier_ID FK from Supplier

(Product_ID, Supplier_ID) Component key for 'Prod_supp' entity]

- Supplier (**Supplier_ID**, Manager_initial, Start_date, End_date, Email)
- Local_supplier (**Supplier_ID** PK/FK, Store_name, City)
- Regional_supplier (**Supplier_ID** PK/FK, Region_name, Vendor_name)

Specialization

Here, Supplier entity is specialized into two subtypes: Local_supplier and Regional_supplier. Both have SupplierID as PK/FK. Local_supplier entity contains Store_name and City while Regional_supplier entity contains Region_name and Vendor_name attributes. This specialization is disjoint as a Supplier can be either Local or Regional supplier but not both.

Normalization:

1NF Rule:

1. Each column must contain atomic values
2. Each column must have a unique name

A user may have multiple phone numbers, then it violates 1NF (not atomic)

- User (**User_ID**, First_name, Last_name, User_email)
- User_Phonenumner (**User_ID** , Phone_number)

Now, there's no atomic values and each column has a unique name. **1NF is satisfied.**

2NF Rule:

1. Must be in 1NF
2. No partial dependency

No partial dependency on composite keys. **2NF is satisfied.**

3NF Rule:

1. Must be in 2NF
2. No transitive dependency

There is no transitive dependency. **3NF is satisfied.**

BCNF Rule:

1. Must be in 3NF
2. No functional dependencies

Every determinant is a candidate key and all relations follow proper key dependency thus, **BCNF is Satisfied.**

4NF Rule:

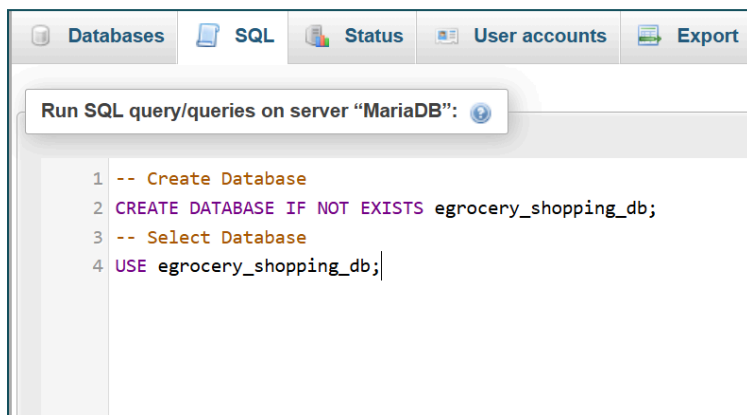
1. Must be in 3NF/BCNF
2. No multi-valued dependency

In this model, Supplier entity is splitted separately into Local and regional sub-entities so local supplier data is separated as well as regional supplier data is separated. No reputation between them.

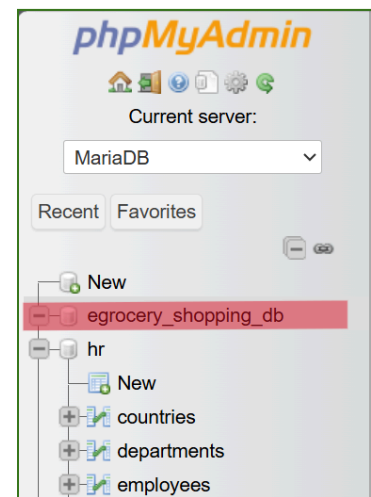
So, no multivalued dependency is this model. **4NF is satisfied.**

SQL implementation

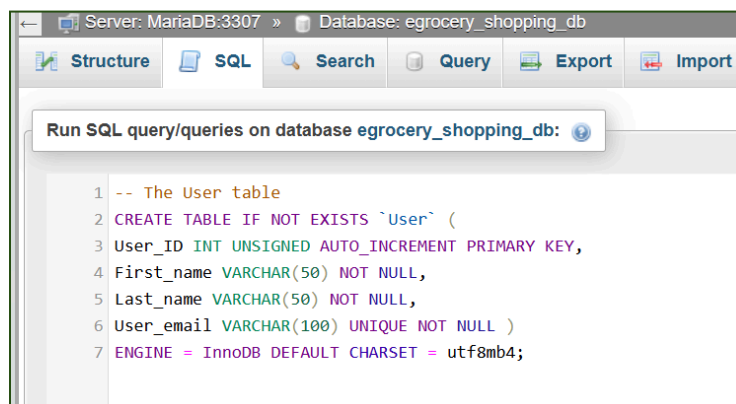
1. Database Creation and Select:



```
1 -- Create Database
2 CREATE DATABASE IF NOT EXISTS egrocery_shopping_db;
3 -- Select Database
4 USE egrocery_shopping_db;
```



2. 'User' Table Creation:



```
1 -- The User table
2 CREATE TABLE IF NOT EXISTS `User` (
3 User_ID INT UNSIGNED AUTO_INCREMENT PRIMARY KEY,
4 First_name VARCHAR(50) NOT NULL,
5 Last_name VARCHAR(50) NOT NULL,
6 User_email VARCHAR(100) UNIQUE NOT NULL )
7 ENGINE = InnoDB DEFAULT CHARSET = utf8mb4;
```

3.Insertion Records in User table:

```
Server: MariaDB:3307 » Database: egrocery_shopping_db » Table: user

Run SQL query/queries on table egrocery_shopping_db.user:

1 -- Insert records in User table
2 INSERT INTO User (First_name, Last_name, User_email)
3 VALUES
4 ('Tasmia', 'Rahman', 'trahman006@gmail.com'),
5 ('Arif', 'Hossain', 'arif.h@gmail.com'),
6 ('Sara', 'Islam', 'sara_islam@gmail.com'),
7 ('Tanvir', 'Ahmed', 'iamtanveer44@yahoo.com'),
8 ('Nusrat', 'Jahan', 'nusrat.j@yahoo.com'),
9 ('Fahim', 'Muntasir', 'fahim.m@gmail.com'),
10 ('Sadia', 'Afrin', 'sadia.afri123n@example.com'),
11 ('Rakib', 'Hasan', 'rakib.h88@gmail.com'),
12 ('Anika', 'Tabassum', 'anika.tab2@gmail.com'),
13 ('Siam', 'Sarker', 'siam.engr@gmail.com'),
14 ('Mehedi', 'Hasan', 'mehedi.ishere@gmail.com'),
15 ('Farhana', 'Akter', 'farhana.akter@gmail.com'),
16 ('Mahmud', 'Khan', 'mahmud.khan99@gmail.com'),
17 ('Ishrat', 'Zaman', 'ishrat.zoe@gmail.com'),
18 ('Tamim', 'Iqbal', 'tamim.iqbal@yahoo.com'),
19 ('Laila', 'Nur', 'laila.nurin67@gmail.com'),
20 ('Sajid', 'Rahman', 'sajid.r.rahman@gmail.com'),
21 ('Nabila', 'Chowdhury', 'nabilaishere.007@gmail.com'),
22 ('Kamrul', 'Islam', 'kamrulislambd@yahoo.com'),
23 ('Sumaiya', 'Anjum', 'sumaiya.a77@gmail.com');
```

	User_ID	First_name	Last_name	User_email
☐	1	Tasmia	Rahman	trahman006@gmail.com
☐	2	Arif	Hossain	arif.h@gmail.com
☐	3	Sara	Islam	sara_islam@gmail.com
☐	4	Tanvir	Ahmed	iamtanveer44@yahoo.com
☐	5	Nusrat	Jahan	nusrat.j@yahoo.com
☐	6	Fahim	Muntasir	fahim.m@gmail.com
☐	7	Sadia	Afrin	sadia.afri123n@example.com
☐	8	Rakib	Hasan	rakib.h88@gmail.com
☐	9	Anika	Tabassum	anika.tab2@gmail.com
☐	10	Siam	Sarker	siam.engr@gmail.com
☐	11	Mehedi	Hasan	mehedi.ishere@gmail.com
☐	12	Farhana	Akter	farhana.akter@gmail.com
☐	13	Mahmud	Khan	mahmud.khan99@gmail.com
☐	14	Ishrat	Zaman	ishrat.zoe@gmail.com
☐	15	Tamim	Iqbal	tamim.iqbal@yahoo.com
☐	16	Laila	Nur	laila.nurin67@gmail.com
☐	17	Sajid	Rahman	sajid.r.rahman@gmail.com
☐	18	Nabila	Chowdhury	nabilaishere.007@gmail.com
☐	19	Kamrul	Islam	kamrulislambd@yahoo.com
☐	20	Sumaiya	Anjum	sumaiya.a77@gmail.com

4. Creation & Insertion of `Reward` Table:

```
Structure SQL Search Query Export Import

Run SQL query/queries on database egrocery_shopping_db:

1 -- Create reward table
2 CREATE TABLE IF NOT EXISTS `Reward` (
3 Reward_ID INT UNSIGNED PRIMARY KEY,
4 User_ID INT UNSIGNED NOT NULL,
5 Reward_name VARCHAR(50) NOT NULL,
6 points INT UNSIGNED NOT NULL,
7 CONSTRAINT fk_user
8 FOREIGN KEY (User_ID)
9 REFERENCES `User` (User_ID)
10 ON DELETE CASCADE )
11 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;
```



```

1  -- Insertion in reward table
2  INSERT INTO `Reward` (`Reward_ID`, `User_ID`, `Reward_name`, `points`)
3  VALUES ('01', 5, 'Cashback Reward', 80),
4  ('02', 12, 'Welcome Bonus', 50),
5  ('03', 3, 'Referral Bonus', 150),
6  ('04', 8, 'Purchase Reward', 100),
7  ('05', 19, 'Loyalty Bonus', 200),
8  ('06', 1, 'Welcome Bonus', 50),
9  ('07', 14, 'Cashback Reward', 80),
10 ('08', 6, 'Purchase Reward', 100),
11 ('09', 10, 'Referral Bonus', 150),
12 ('10', 17, 'Welcome Bonus', 50),
13 ('11', 2, 'Loyalty Bonus', 200),
14 ('12', 9, 'Purchase Reward', 100),
15 ('13', 15, 'Cashback Reward', 80),
16 ('14', 4, 'Welcome Bonus', 50),
17 ('15', 20, 'Referral Bonus', 150),
18 ('16', 7, 'Loyalty Bonus', 200),
19 ('17', 11, 'Cashback Reward', 80),
20 ('18', 13, 'Purchase Reward', 100),
21 ('19', 18, 'Welcome Bonus', 50),
22 ('20', 16, 'Referral Bonus', 150),
23 ('21', 5, 'Welcome Bonus', 50),
24 ('22', 3, 'Cashback Reward', 80),
25 ('23', 8, 'Referral Bonus', 150),
26 ('24', 12, 'Purchase Reward', 100),
27 ('25', 1, 'Loyalty Bonus', 200) ;

```

			Reward_ID	User_ID	Reward_name	points
☐	Edit	Copy	Delete		6 Cashback Reward	80
☐	Edit	Copy	Delete		7 Welcome Bonus	50
☐	Edit	Copy	Delete	3	3 Referral Bonus	150
☐	Edit	Copy	Delete	4	8 Purchase Reward	100
☐	Edit	Copy	Delete	5	19 Loyalty Bonus	200
☐	Edit	Copy	Delete	6	1 Welcome Bonus	50
☐	Edit	Copy	Delete	7	14 Cashback Reward	80
☐	Edit	Copy	Delete	8	6 Purchase Reward	100
☐	Edit	Copy	Delete	9	10 Referral Bonus	150
☐	Edit	Copy	Delete	10	17 Welcome Bonus	50
☐	Edit	Copy	Delete	11	2 Loyalty Bonus	200
☐	Edit	Copy	Delete	12	9 Purchase Reward	100
☐	Edit	Copy	Delete	13	15 Cashback Reward	80
☐	Edit	Copy	Delete	14	4 Welcome Bonus	50
☐	Edit	Copy	Delete	15	20 Referral Bonus	150
☐	Edit	Copy	Delete	16	7 Loyalty Bonus	200
☐	Edit	Copy	Delete	17	11 Cashback Reward	80
☐	Edit	Copy	Delete	18	13 Purchase Reward	100
☐	Edit	Copy	Delete	19	18 Welcome Bonus	50
☐	Edit	Copy	Delete	20	16 Referral Bonus	150
☐	Edit	Copy	Delete	21	5 Welcome Bonus	50
☐	Edit	Copy	Delete	22	3 Cashback Reward	80
☐	Edit	Copy	Delete	23	8 Referral Bonus	150
☐	Edit	Copy	Delete	24	12 Purchase Reward	100
☐	Edit	Copy	Delete	25	1 Loyalty Bonus	200

5. Creation & Insertion of `Cart` Table:

Run SQL query/queries on database egrocery_shopping_db: ?

```
1 -- Cart table creation
2 CREATE TABLE IF NOT EXISTS `Cart` (
3   Cart_ID VARCHAR(10) PRIMARY KEY,
4   User_ID INT UNSIGNED NOT NULL,
5   Quantity INT UNSIGNED NOT NULL,
6   Status VARCHAR(20) NOT NULL,
7   CONSTRAINT fk_cart_user
8   FOREIGN KEY (User_ID)
9   REFERENCES `User` (User_ID)
10  ON DELETE CASCADE)
11 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;
```

Run SQL query/queries on table egrocery_shopping_db.cart: ?

```

1 INSERT INTO Cart (Cart_ID, User_ID, Quantity, Status)
2 VALUES ('A001', 7, 4, 'Active'),
3 ('A002', 3, 1, 'Ordered'),
4 ('A003', 12, 4, 'Active'),
5 ('A004', 5, 2, 'Cancelled'),
6 ('A005', 1, 8, 'Active'),
7 ('A006', 9, 6, 'Ordered'),
8 ('A007', 14, 5, 'Active'),
9 ('A008', 2, 1, 'Cancelled'),
10 ('A009', 10, 3, 'Active'),
11 ('A010', 6, 10, 'Ordered'),
12 ('A011', 15, 2, 'Active'),
13 ('A012', 8, 4, 'Ordered'),
14 ('A013', 4, 1, 'Cancelled'),
15 ('A014', 11, 3, 'Active'),
16 ('A015', 20, 2, 'Ordered'),
17 ('A016', 13, 5, 'Active'),
18 ('A017', 18, 2, 'Cancelled'),
19 ('A018', 16, 10, 'Active'),
20 ('A019', 19, 9, 'Ordered'),
21 ('A020', 17, 2, 'Active'),
22 ('A021', 1, 4, 'Active'),
23 ('A022', 4, 2, 'Active'),
24 ('A023', 11, 10, 'Cancelled'),
25 ('A024', 9, 3, 'Ordered'),
26 ('A025', 20, 2, 'Cancelled');

```

	Cart_ID	User_ID	Quantity	Status
☐ Edit ☐ Copy ☐ Delete	A001	7	4	Active
☐ Edit ☐ Copy ☐ Delete	A002	3	1	Ordered
☐ Edit ☐ Copy ☐ Delete	A003	12	4	Active
☐ Edit ☐ Copy ☐ Delete	A004	5	2	Cancelled
☐ Edit ☐ Copy ☐ Delete	A005	1	8	Active
☐ Edit ☐ Copy ☐ Delete	A006	9	6	Ordered
☐ Edit ☐ Copy ☐ Delete	A007	14	5	Active
☐ Edit ☐ Copy ☐ Delete	A008	2	1	Cancelled
☐ Edit ☐ Copy ☐ Delete	A009	10	3	Active
☐ Edit ☐ Copy ☐ Delete	A010	6	10	Ordered
☐ Edit ☐ Copy ☐ Delete	A011	15	2	Active
☐ Edit ☐ Copy ☐ Delete	A012	8	4	Ordered
☐ Edit ☐ Copy ☐ Delete	A013	4	1	Cancelled
☐ Edit ☐ Copy ☐ Delete	A014	11	3	Active
☐ Edit ☐ Copy ☐ Delete	A015	20	2	Ordered
☐ Edit ☐ Copy ☐ Delete	A016	13	5	Active
☐ Edit ☐ Copy ☐ Delete	A017	18	2	Cancelled
☐ Edit ☐ Copy ☐ Delete	A018	16	10	Active
☐ Edit ☐ Copy ☐ Delete	A019	19	9	Ordered
☐ Edit ☐ Copy ☐ Delete	A020	17	2	Active
☐ Edit ☐ Copy ☐ Delete	A021	1	4	Active
☐ Edit ☐ Copy ☐ Delete	A022	4	2	Active
☐ Edit ☐ Copy ☐ Delete	A023	11	10	Cancelled

6. Creation & Insertion of `Order` Table:

Structure SQL Search Query Export

Run SQL query/queries on database egrocery_shopping_db: ?

```

1 -- Order table create
2 CREATE TABLE `Order` (
3 Order_ID VARCHAR(10) PRIMARY KEY,
4 User_ID INT UNSIGNED NOT NULL,
5 Order_Date DATE NOT NULL,
6 Type VARCHAR(20),
7 FOREIGN KEY (User_ID)
8 REFERENCES User(User_ID)
9 ON DELETE CASCADE ON UPDATE CASCADE )
10 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;

```

Run SQL query/queries on table egrocery_shopping_db.order: ⓘ

```

1 INSERT INTO `Order` (Order_ID, User_ID, Order_Date, Type)
2 VALUES ('001', 3, '2026-01-05', 'Online'),
3 ('002', 7, '2026-01-06', 'COD'),
4 ('003', 1, '2026-01-07', 'COD'),
5 ('004', 5, '2026-01-08', 'COD'),
6 ('005', 3, '2026-01-09', 'COD'),
7 ('006', 9, '2026-01-10', 'COD'),
8 ('007', 1, '2026-01-11', 'Online'),
9 ('008', 4, '2026-01-12', 'COD'),
10 ('009', 7, '2026-02-13', 'Online'),
11 ('010', 2, '2026-02-14', 'COD'),
12 ('011', 6, '2026-02-15', 'Online'),
13 ('012', 8, '2026-02-16', 'COD'),
14 ('013', 3, '2026-03-17', 'Online'),
15 ('014', 10, '2026-03-18', 'COD'),
16 ('015', 5, '2026-03-19', 'COD'),
17 ('016', 1, '2026-03-20', 'COD'),
18 ('017', 2, '2026-03-21', 'Online'),
19 ('018', 9, '2026-03-22', 'COD'),
20 ('019', 7, '2026-03-23', 'COD'),
21 ('020', 4, '2026-03-24', 'COD'),
22 ('021', 4, '2026-04-14', 'COD'),
23 ('022', 4, '2026-04-10', 'Online') ;

```

	Order_ID	User_ID	Order_Date	Type
☐ Edit ☐ Copy ☐ Delete	001	3	2026-01-05	Online
☐ Edit ☐ Copy ☐ Delete	002	7	2026-01-06	COD
☐ Edit ☐ Copy ☐ Delete	003	1	2026-01-07	COD
☐ Edit ☐ Copy ☐ Delete	004	5	2026-01-08	COD
☐ Edit ☐ Copy ☐ Delete	005	3	2026-01-09	COD
☐ Edit ☐ Copy ☐ Delete	006	9	2026-01-10	COD
☐ Edit ☐ Copy ☐ Delete	007	1	2026-01-11	Online
☐ Edit ☐ Copy ☐ Delete	008	4	2026-01-12	COD
☐ Edit ☐ Copy ☐ Delete	009	7	2026-02-13	Online
☐ Edit ☐ Copy ☐ Delete	010	2	2026-02-14	COD
☐ Edit ☐ Copy ☐ Delete	011	6	2026-02-15	Online
☐ Edit ☐ Copy ☐ Delete	012	8	2026-02-16	COD
☐ Edit ☐ Copy ☐ Delete	013	3	2026-03-17	Online
☐ Edit ☐ Copy ☐ Delete	014	10	2026-03-18	COD
☐ Edit ☐ Copy ☐ Delete	015	5	2026-03-19	COD
☐ Edit ☐ Copy ☐ Delete	016	1	2026-03-20	COD
☐ Edit ☐ Copy ☐ Delete	017	2	2026-03-21	Online
☐ Edit ☐ Copy ☐ Delete	018	9	2026-03-22	COD
☐ Edit ☐ Copy ☐ Delete	019	7	2026-03-23	COD
☐ Edit ☐ Copy ☐ Delete	020	4	2026-03-24	COD
☐ Edit ☐ Copy ☐ Delete	021	4	2026-04-14	COD
☐ Edit ☐ Copy ☐ Delete	022	4	2026-04-10	Online

Console

7. Creation & Insertion of `Supplier` Table:

Run SQL query/queries on database egrocery_shopping_db: ⓘ

```

1 CREATE TABLE `Supplier` (
2 Supplier_ID INT UNSIGNED AUTO_INCREMENT NOT NULL,
3 Manager_initial VARCHAR(10) NOT NULL,
4 Start_date DATE NOT NULL,
5 End_date DATE, -- can be NULL
6 Email VARCHAR(100) UNIQUE NOT NULL,
7 PRIMARY KEY (Supplier_ID) )
8 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;

```

Run SQL query/queries on table egrocery_shopping_db.supplier: ⓘ

```

1 -- Insertion record in Supplier table
2 INSERT INTO Supplier (Supplier_ID, Manager_initial, Start_date, End_date, Email)
3 VALUES
4 (1, 'ASR', '2025-12-30', NULL, 'tatkab@gmail.com'),
5 (2, 'MRA', '2025-12-28', NULL, 'allfresh@yahoo.com'),
6 (3, 'TK', '2025-12-30', '2026-01-30', 'bdbazar@gmail.com'),
7 (4, 'NH', '2025-12-29', NULL, 'freshcook@gmail.com'),
8 (5, 'RB', '2025-12-30', NULL, 'allgood66@yahoo.com'),
9 (6, 'SAM', '2026-01-20', NULL, 'chaldal@gmail.com'),
10 (7, 'KIA', '2026-01-25', NULL, 'ghorbazar55@gmail.com'),
11 (8, 'FR', '2026-01-17', '2026-01-27', 'bdbazar55@gmail.com'),
12 (9, 'JRD', '2026-02-05', NULL, 'supplybd@gmail.com'),
13 (10, 'LMM', '2026-02-12', NULL, 'supplydhk@gmail.com') ;

```

Extra options						
← T →						
		Supplier_ID	Manager_initial	Start_date	End_date	Email
☐	Edit Copy Delete	1	ASR	2025-12-30	NULL	tatkabd@gmail.com
☐	Edit Copy Delete	2	MRA	2025-12-28	NULL	allfresh@yahoo.com
☐	Edit Copy Delete	3	TK	2025-12-30	2026-01-30	bdbazar@gmail.com
☐	Edit Copy Delete	4	NH	2025-12-29	NULL	freshcook@gmail.com
☐	Edit Copy Delete	5	RB	2025-12-30	NULL	allgood66@yahoo.com
☐	Edit Copy Delete	6	SAM	2026-01-20	NULL	chaldal@gmail.com
☐	Edit Copy Delete	7	KIA	2026-01-25	NULL	ghorbazar55@gmail.com
☐	Edit Copy Delete	8	FR	2026-01-17	2026-01-27	bdbazar55@gmail.com
☐	Edit Copy Delete	9	JRD	2026-02-05	NULL	supplybd@gmail.com
☐	Edit Copy Delete	10	LMM	2026-02-12	NULL	supplydhk@gmail.com
☐ Check all With selected: Edit Copy Delete Export						

8. Creation & Insertion of `Local\_supplier` Table:

```

1 INSERT INTO Local_Supplier (Supplier_ID, Store_Name, City)
2 VALUES
3 (1, 'Tatka Mart BD', 'Dhaka'),
4 (4, 'Fresh Cook', 'Dhaka'),
5 (7, 'Ghor Bazar Shop', 'Dhaka'),
6 (8, 'BD Bazar.com', 'Savar'),
7 (9, 'Supply BD Basket', 'Savar'),
8 (10, 'Prime Grocery Point', 'Dhaka') ;

```

Extra options			
← T →			
	Supplier_ID	Store_Name	City
☐	1	Tatka Mart BD	Dhaka
☐	4	Fresh Cook	Dhaka
☐	7	Ghor Bazar Shop	Dhaka
☐	8	BD Bazar.com	Savar
☐	9	Supply BD Basket	Savar
☐	10	Prime Grocery Point	Dhaka
☐ Check all With selected: Edit Copy Delete			

Run SQL query/queries on database egrocery_shopping_db: ?

```

1 CREATE TABLE Local_Supplier (
2 Supplier_ID INT UNSIGNED PRIMARY KEY,
3 Store_Name VARCHAR(100) NOT NULL,
4 City VARCHAR(50) NOT NULL,
5 FOREIGN KEY (Supplier_ID)
6 REFERENCES Supplier(Supplier_ID)
7 ON DELETE CASCADE ON UPDATE CASCADE)
8 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;

```

9. Creation & Insertion of `Regional\_supplier` Table:

Run SQL query/queries on database egrocery_shopping_db: ?

```

1 CREATE TABLE Regional_Supplier (
2 Supplier_ID INT UNSIGNED PRIMARY KEY,
3 Region_name VARCHAR(100) NOT NULL,
4 Vendor_name VARCHAR(100) NOT NULL,
5 FOREIGN KEY (Supplier_ID) -- Foreign key from Supplier table
6 REFERENCES Supplier(Supplier_ID)
7 ON DELETE CASCADE ON UPDATE CASCADE)
8 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4;

```

Run SQL query/queries on table egrocery_shopping_db.regional_supplier: ?

```

1 INSERT INTO Regional_Supplier (Supplier_ID, Region_name, Vendor_name)
2 VALUES
3 (2, 'Chittagong Division', 'Bay Bengal CTG'),
4 (5, 'Sylhet Division', 'Tea Garden LTD.');
```

	Supplier_ID	Region_name	Vendor_name
☐ Edit Copy Delete	2	Chittagong Division	Bay Bengal CTG
☐ Edit Copy Delete	5	Sylhet Division	Tea Garden LTD.

☐ Check all
 With selected:
 ☐ Edit
 ☐ Copy
 ☐ Delete
 ☐ Export

10. Creation & Insertion of `Product` Table:

Run SQL query/queries on database egrocery_shopping_db:

```
1 CREATE TABLE IF NOT EXISTS Product (  
2 Product_ID VARCHAR(10) NOT NULL,  
3 Supplier_ID INT UNSIGNED NOT NULL,  
4 Product_name VARCHAR(100) NOT NULL,  
5 Price DECIMAL(10,2) NOT NULL,  
6 Category VARCHAR(50) NOT NULL,  
7 PRIMARY KEY (Product_ID), FOREIGN KEY (Supplier_ID)  
8 REFERENCES Supplier(Supplier_ID)  
9 ON DELETE CASCADE ON UPDATE CASCADE)  
10 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;
```

	Product_ID	Supplier_ID	Product_name	Price	Category
☐ Edit ☐ Copy ☐ Delete	P001	1	Rice 5kg	520.00	Grocery
☐ Edit ☐ Copy ☐ Delete	P002	2	Wheat Flour 2kg	180.00	Grocery
☐ Edit ☐ Copy ☐ Delete	P003	5	Sugar 1kg	120.00	Grocery
☐ Edit ☐ Copy ☐ Delete	P004	4	Salt 1kg	45.00	Grocery
☐ Edit ☐ Copy ☐ Delete	P005	5	Cooking Oil 1L	220.00	Grocery
☐ Edit ☐ Copy ☐ Delete	P006	3	Milk Powder 500g	340.00	Dairy
☐ Edit ☐ Copy ☐ Delete	P007	7	Butter 200g	160.00	Dairy
☐ Edit ☐ Copy ☐ Delete	P008	3	Cheese 250g	280.00	Dairy
☐ Edit ☐ Copy ☐ Delete	P009	9	Yogurt 500g	90.00	Dairy
☐ Edit ☐ Copy ☐ Delete	P010	10	Eggs 12 pcs	150.00	Dairy
☐ Edit ☐ Copy ☐ Delete	P011	1	Soap Pack	110.00	Personal Care
☐ Edit ☐ Copy ☐ Delete	P012	2	Shampoo 200ml	180.00	Personal Care
☐ Edit ☐ Copy ☐ Delete	P013	2	Toothpaste	95.00	Personal Care
☐ Edit ☐ Copy ☐ Delete	P014	4	Toothbrush Set	120.00	Personal Care
☐ Edit ☐ Copy ☐ Delete	P015	5	Face Wash	250.00	Personal Care
☐ Edit ☐ Copy ☐ Delete	P016	6	Biscuit Pack	60.00	Snacks
☐ Edit ☐ Copy ☐ Delete	P017	3	Chips Pack	40.00	Snacks
☐ Edit ☐ Copy ☐ Delete	P018	1	Chocolate Bar	90.00	Snacks
☐ Edit ☐ Copy ☐ Delete	P019	8	Cold Drink 1L	70.00	Beverage
☐ Edit ☐ Copy ☐ Delete	P020	10	Juice 1L	130.00	Beverage
☐ Edit ☐ Copy ☐ Delete	P021	1	Green Tea Box	200.00	Beverage

```
1 INSERT INTO Product (Product_ID, Supplier_ID, Product_name, Price, Category)  
2 VALUES  
3 ('P001', 1, 'Rice 5kg', 520.00, 'Grocery'),  
4 ('P002', 2, 'Wheat Flour 2kg', 180.00, 'Grocery'),  
5 ('P003', 5, 'Sugar 1kg', 120.00, 'Grocery'),  
6 ('P004', 4, 'Salt 1kg', 45.00, 'Grocery'),  
7 ('P005', 5, 'Cooking Oil 1L', 220.00, 'Grocery'),  
8 ('P006', 3, 'Milk Powder 500g', 340.00, 'Dairy'),  
9 ('P007', 7, 'Butter 200g', 160.00, 'Dairy'),  
10 ('P008', 3, 'Cheese 250g', 280.00, 'Dairy'),  
11 ('P009', 9, 'Yogurt 500g', 90.00, 'Dairy'),  
12 ('P010', 10, 'Eggs 12 pcs', 150.00, 'Dairy'),  
13 ('P011', 1, 'Soap Pack', 110.00, 'Personal Care'),  
14 ('P012', 2, 'Shampoo 200ml', 180.00, 'Personal Care'),  
15 ('P013', 2, 'Toothpaste', 95.00, 'Personal Care'),  
16 ('P014', 4, 'Toothbrush Set', 120.00, 'Personal Care'),  
17 ('P015', 5, 'Face Wash', 250.00, 'Personal Care'),  
18 ('P016', 6, 'Biscuit Pack', 60.00, 'Snacks'),  
19 ('P017', 3, 'Chips Pack', 40.00, 'Snacks'),  
20 ('P018', 1, 'Chocolate Bar', 90.00, 'Snacks'),  
21 ('P019', 8, 'Cold Drink 1L', 70.00, 'Beverage'),  
22 ('P020', 10, 'Juice 1L', 130.00, 'Beverage'),  
23 ('P021', 1, 'Green Tea Box', 200.00, 'Beverage'),  
24 ('P022', 2, 'Coffee Jar', 450.00, 'Beverage'),  
25 ('P023', 9, 'Hand Wash', 160.00, 'Hygiene'),  
26 ('P024', 4, 'Detergent Powder', 300.00, 'Cleaning'),  
27 ('P025', 5, 'Dish Wash Liquid', 140.00, 'Cleaning'),  
28 ('P026', 6, 'Floor Cleaner', 220.00, 'Cleaning'),
```

11. Creation & Insertion of `Payment` Table:

Run SQL query/queries on database egrocery_shopping_db:

```
1 CREATE TABLE IF NOT EXISTS Payment (  
2 Order_ID VARCHAR(10) NOT NULL,  
3 Payment_ID VARCHAR(10) NOT NULL,  
4 Amount DECIMAL(10,2) NOT NULL,  
5 Pay_date DATE NOT NULL,  
6 Method VARCHAR(20) NOT NULL,  
7 PRIMARY KEY (Order_ID, Payment_ID),  
8 FOREIGN KEY (Order_ID)  
9 REFERENCES `Order` (Order_ID)  
10 ON DELETE CASCADE ON UPDATE CASCADE )  
11 ENGINE=InnoDB  
12 DEFAULT CHARSET=utf8mb4 ;
```

```

1 INSERT INTO Payment (Order_ID, Payment_ID, Amount, Pay_date, Method)
2 VALUES
3 ('001', 'P001', 520.00, '2026-01-05', 'Online'),
4 ('002', 'P002', 180.00, '2026-01-06', 'COD'),
5 ('003', 'P003', 120.00, '2026-01-07', 'COD'),
6 ('004', 'P004', 45.00, '2026-01-08', 'COD'),
7 ('005', 'P005', 220.00, '2026-01-09', 'COD'),
8 ('006', 'P006', 340.00, '2026-01-10', 'COD'),
9 ('007', 'P007', 160.00, '2026-01-11', 'Online'),
10 ('008', 'P008', 280.00, '2026-01-12', 'COD'),
11 ('009', 'P009', 90.00, '2026-02-13', 'Online'),
12 ('010', 'P010', 150.00, '2026-02-14', 'COD'),
13 ('011', 'P011', 450.00, '2026-02-15', 'Online'),
14 ('012', 'P012', 80.00, '2026-02-16', 'COD'),
15 ('013', 'P013', 520.00, '2026-03-17', 'Online'),
16 ('014', 'P014', 600.00, '2026-03-18', 'COD'),
17 ('015', 'P015', 140.00, '2026-03-19', 'COD'),
18 ('016', 'P016', 50.00, '2026-03-20', 'COD'),
19 ('017', 'P017', 110.00, '2026-03-21', 'Online'),
20 ('018', 'P018', 260.00, '2026-03-22', 'COD'),
21 ('019', 'P019', 70.00, '2026-03-23', 'COD'),
22 ('020', 'P020', 300.00, '2026-03-24', 'COD'),
23 ('021', 'P021', 220.00, '2026-04-14', 'COD'),
24 ('022', 'P022', 250.00, '2026-04-14', 'Online');

```

	Order_ID	Payment_ID	Amount	Pay_date	Method
☐	001	P001	520.00	2026-01-05	Online
☐	002	P002	180.00	2026-01-06	COD
☐	003	P003	120.00	2026-01-07	COD
☐	004	P004	45.00	2026-01-08	COD
☐	005	P005	220.00	2026-01-09	COD
☐	006	P006	340.00	2026-01-10	COD
☐	007	P007	160.00	2026-01-11	Online
☐	008	P008	280.00	2026-01-12	COD
☐	009	P009	90.00	2026-02-13	Online
☐	010	P010	150.00	2026-02-14	COD
☐	011	P011	450.00	2026-02-15	Online
☐	012	P012	80.00	2026-02-16	COD
☐	013	P013	520.00	2026-03-17	Online
☐	014	P014	600.00	2026-03-18	COD
☐	015	P015	140.00	2026-03-19	COD
☐	016	P016	50.00	2026-03-20	COD
☐	017	P017	110.00	2026-03-21	Online
☐	018	P018	260.00	2026-03-22	COD
☐	019	P019	70.00	2026-03-23	COD
☐	020	P020	300.00	2026-03-24	COD
☐	021	P021	220.00	2026-04-14	COD
☐	022	P022	250.00	2026-04-14	Online

12. Creation & Insertion of `Cart\_prod` Table:

Run SQL query/queries on database `egrocery_shopping_db`:

```

1 -- Juntion table for Cart and Product table
2 CREATE TABLE `Cart_prod` (
3   Cart_ID VARCHAR(10) NOT NULL,
4   Product_ID VARCHAR(10) NOT NULL,
5   PRIMARY KEY (Cart_ID, Product_ID), -- Composite key
6   FOREIGN KEY (Cart_ID) REFERENCES `Cart` (Cart_ID)
7   ON DELETE CASCADE ON UPDATE CASCADE,
8   FOREIGN KEY (Product_ID) REFERENCES `Product` (Product_ID)
9   ON DELETE CASCADE ON UPDATE CASCADE)
10 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;

```


		Cart_ID	Product_ID
☐	Edit	A001	P001
☐	Edit	A006	P002
☐	Edit	A003	P003
☐	Edit	A007	P004
☐	Edit	A002	P005
☐	Edit	A008	P006
☐	Edit	A009	P007
☐	Edit	A010	P008
☐	Edit	A011	P009
☐	Edit	A004	P010
☐	Edit	A004	P011
☐	Edit	A012	P012
☐	Edit	A013	P013
☐	Edit	A014	P014
☐	Edit	A001	P015
☐	Edit	A015	P016
☐	Edit	A016	P017
☐	Edit	A017	P018
☐	Edit	A018	P019
☐	Edit	A005	P020

```
Run SQL query/queries on table egrocery_shopping_db.cart_prod:
1 INSERT INTO `Cart_prod` (Cart_ID, Product_ID)
2 VALUES
3 ('A001', 'P001'), ('A001', 'P015'),
4 ('A001', 'P030'), ('A002', 'P005'),
5 ('A002', 'P022'), ('A003', 'P003'),
6 ('A003', 'P040'), ('A004', 'P010'),
7 ('A004', 'P011'), ('A005', 'P020'),
8 ('A005', 'P025'), ('A006', 'P002'),
9 ('A007', 'P004'), ('A008', 'P006'),
10 ('A009', 'P007'), ('A010', 'P008'),
11 ('A011', 'P009'), ('A012', 'P012'),
12 ('A013', 'P013'), ('A014', 'P014'),
13 ('A015', 'P016'), ('A016', 'P017'),
14 ('A017', 'P018'), ('A018', 'P019'),
15 ('A019', 'P021'), ('A020', 'P023'),
16 ('A021', 'P024'), ('A022', 'P026'),
17 ('A023', 'P027'), ('A024', 'P028');
```

13. Creation & Insertion of `Orer\_prod` Table:

```
Run SQL query/queries on database egrocery_shopping_db:
1 -- Junction table Order_prod
2 CREATE TABLE `Order_prod` (
3 Order_ID VARCHAR(10) NOT NULL,
4 Product_ID VARCHAR(10) NOT NULL,
5 PRIMARY KEY (Order_ID, Product_ID), -- Component key
6 FOREIGN KEY (Order_ID) REFERENCES `Order` (Order_ID)
7 ON DELETE CASCADE ON UPDATE CASCADE,
8 FOREIGN KEY (Product_ID) REFERENCES `Product` (Product_ID)
9 ON DELETE CASCADE ON UPDATE CASCADE
10 ) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 |;
```


Run SQL query/queries on table `egrocery_shopping_db.order_pro`

```

1 INSERT INTO `Order_prod` (Order_ID, Product_ID)
2 VALUES
3 ('001', 'P001'), ('001', 'P002'),
4 ('001', 'P010'), ('002', 'P005'),
5 ('002', 'P020'), ('003', 'P025'), ('003', 'P015'),
6 ('004', 'P003'), ('005', 'P004'), ('006', 'P006'),
7 ('007', 'P007'), ('008', 'P008'), ('009', 'P009'),
8 ('010', 'P011'), ('011', 'P012'), ('012', 'P013'),
9 ('013', 'P014'), ('014', 'P016'), ('015', 'P017'),
10 ('016', 'P018'), ('017', 'P019'), ('018', 'P021'),
11 ('019', 'P022'), ('020', 'P023'), ('021', 'P024'),
12 ('022', 'P040');

```

	Order_ID	Product_ID
☐ Edit Copy Delete	001	P001
☐ Edit Copy Delete	001	P002
☐ Edit Copy Delete	004	P003
☐ Edit Copy Delete	005	P004
☐ Edit Copy Delete	002	P005
☐ Edit Copy Delete	006	P006
☐ Edit Copy Delete	007	P007
☐ Edit Copy Delete	008	P008
☐ Edit Copy Delete	009	P009
☐ Edit Copy Delete	001	P010
☐ Edit Copy Delete	010	P011
☐ Edit Copy Delete	011	P012
☐ Edit Copy Delete	012	P013
☐ Edit Copy Delete	013	P014
☐ Edit Copy Delete	003	P015
☐ Edit Copy Delete	014	P016
☐ Edit Copy Delete	015	P017
☐ Edit Copy Delete	016	P018
☐ Edit Copy Delete	017	P019
☐ Edit Copy Delete	002	P020

14. Creation & Insertion of `Prod\_supp` Table:

Run SQL query/queries on database `egrocery_shopping_db`:

```

1 CREATE TABLE `Prod_supp` (
2 Product_ID VARCHAR(10) NOT NULL,
3 Supplier_ID INT UNSIGNED NOT NULL,
4 PRIMARY KEY (Product_ID, Supplier_ID),
5 FOREIGN KEY (Product_ID) REFERENCES `Product` (Product_ID)
6 ON DELETE CASCADE ON UPDATE CASCADE,
7 FOREIGN KEY (Supplier_ID) REFERENCES `Supplier` (Supplier_ID)
8 ON DELETE CASCADE ON UPDATE CASCADE )
9 ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 ;

```

```
Run SQL query/queries on table egrocery_shopping_db.prod_supp:

1 INSERT INTO `Prod_supp` (Product_ID, Supplier_ID)
2 VALUES
3 ('P001', 1), ('P002', 1), ('P003', 1), ('P004', 1),
4 ('P005', 2), ('P006', 2), ('P007', 2), ('P008', 2),
5 ('P009', 3), ('P010', 3), ('P011', 3), ('P012', 3),
6 ('P013', 4), ('P014', 4), ('P015', 4), ('P016', 4),
7 ('P017', 5), ('P018', 5), ('P019', 5), ('P020', 5),
8 ('P021', 6), ('P022', 6), ('P023', 6), ('P024', 6),
9 ('P025', 7), ('P026', 7), ('P027', 7), ('P028', 7),
10 ('P029', 8), ('P030', 8), ('P031', 8), ('P032', 8),
11 ('P033', 9), ('P034', 9), ('P035', 9), ('P036', 9),
12 ('P037', 10), ('P038', 10), ('P039', 10), ('P040', 10) ;
```

	Product_ID	Supplier_ID
☐ Edit ☐ Copy ☐ Delete	P001	
☐ Edit ☐ Copy ☐ Delete	P002	
☐ Edit ☐ Copy ☐ Delete	P003	
☐ Edit ☐ Copy ☐ Delete	P004	
☐ Edit ☐ Copy ☐ Delete	P005	
☐ Edit ☐ Copy ☐ Delete	P006	
☐ Edit ☐ Copy ☐ Delete	P007	
☐ Edit ☐ Copy ☐ Delete	P008	
☐ Edit ☐ Copy ☐ Delete	P009	
☐ Edit ☐ Copy ☐ Delete	P010	
☐ Edit ☐ Copy ☐ Delete	P011	
☐ Edit ☐ Copy ☐ Delete	P012	
☐ Edit ☐ Copy ☐ Delete	P013	
☐ Edit ☐ Copy ☐ Delete	P014	
☐ Edit ☐ Copy ☐ Delete	P015	
☐ Edit ☐ Copy ☐ Delete	P016	
☐ Edit ☐ Copy ☐ Delete	P017	

15. `egrocery\_shopping\_bd` Database View:

phpMyAdmin

Current server: MariaDB

Recent Favorites

New

egrocery_shopping_db

- New
- cart
- cart_prod
- local_supplier
- order
- order_prod
- payment
- product
- prod_supp
- regional_supplier
- reward
- supplier
- user
- user_phonenumber

Server: MariaDB:3307 Database: egrocery_shopping_db

Structure SQL Search Query Export Import Operations Privileges Routines Events

Table	Action	Rows	Type	Collation	Size	Over
☐ cart	★ Browse Structure Search Insert Empty Drop	25	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ cart_prod	★ Browse Structure Search Insert Empty Drop	30	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ local_supplier	★ Browse Structure Search Insert Empty Drop	6	InnoDB	utf8mb4_general_ci	16.0 KiB	
☐ order	★ Browse Structure Search Insert Empty Drop	22	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ order_prod	★ Browse Structure Search Insert Empty Drop	26	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ payment	★ Browse Structure Search Insert Empty Drop	22	InnoDB	utf8mb4_general_ci	16.0 KiB	
☐ product	★ Browse Structure Search Insert Empty Drop	40	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ prod_supp	★ Browse Structure Search Insert Empty Drop	40	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ regional_supplier	★ Browse Structure Search Insert Empty Drop	2	InnoDB	utf8mb4_general_ci	16.0 KiB	
☐ reward	★ Browse Structure Search Insert Empty Drop	25	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ supplier	★ Browse Structure Search Insert Empty Drop	10	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ user	★ Browse Structure Search Insert Empty Drop	20	InnoDB	utf8mb4_general_ci	32.0 KiB	
☐ user_phonenumber	★ Browse Structure Search Insert Empty Drop	25	InnoDB	utf8mb4_general_ci	16.0 KiB	
13 tables	Sum	293	MyISAM	utf8mb4_unicode_ci	352.0 KiB	

Here, The `egrocery\_shopping\_bd` database has a total of 13 tables (after normalization).

Some Query from `egrocery\_shopping\_bd` Database:

Using LEFT JOIN

Query 01: Find users who did not receive the “welcome reward”. Show their last name, ID and email address.

```
1 | SELECT U.Last_name, U.User_ID, U.User_email
2 | FROM `User` AS U
3 | LEFT JOIN Reward AS R
4 |     ON U.User_ID = R.User_ID
5 |     AND r.Reward_name = 'Welcome Bonus'
6 | WHERE R.User_ID IS NULL;
```

Last_name	User_ID	User_email
Hossain	2	arif.h@gmail.com
Islam	3	sara_islam@gmail.com
Muntasir	6	fahim.m@gmail.com
Afrin	7	sadia.afri123n@example.com
Hasan	8	rakib.h88@gmail.com
Tabassum	9	anika.tab2@gmail.com
Sarker	10	siam.engr@gmail.com
Hasan	11	mehedi.ishere@gmail.com
Khan	13	mahmud.khan99@gmail.com
Zaman	14	ishrat.zoe@gmail.com
Iqbal	15	tamim.iqbal@yahoo.com
Nur	16	laila.nurin67@gmail.com
Islam	19	kamrulislambd@yahoo.com
Anjum	20	sumaiya.a77@gmail.com

Using INNER JOIN

Query 02: Find local supplier ID and name who are still supplying products.

```
1 SELECT S.Supplier_ID, LS.Store_name
2 FROM supplier AS S
3 INNER JOIN local_supplier AS LS
4 ON S.Supplier_ID = LS.Supplier_ID AND
5 S.End_date IS NULL;
```

Supplier_ID	Store_name
1	Tatka Mart BD
4	Fresh Cook
7	Ghor Bazar Shop
9	Supply BD Basket
10	Prime Grocery Point

Using Inner Join with Conditions

Query 03: Find those User's ID who ordered using Cash on Delivery in the January month including the payment method.

```
1 SELECT O.User_ID , P.Method
2 FROM `order` AS O
3 INNER JOIN payment AS P
4 ON O.Order_ID = P.Order_ID
5 WHERE P.Method = 'COD' AND
6 P.Pay_date BETWEEN '2026-01-01' AND '2026-01-31';
```

Extra options

User_ID	Method
1	COD
3	COD
4	COD
5	COD
7	COD
9	COD

Simple Query Example

Query 04: Find those product name and ID that is 'beverage' and price less than 200.

```
1 SELECT Product_ID, Product_name
2 FROM `product`
3 WHERE Category = 'Beverage' AND Price < 200;
```

				Product_ID	Product_name
☐	 Edit	 Copy	 Delete	P019	Cold Drink 1L
☐	 Edit	 Copy	 Delete	P020	Juice 1L
☐	 Edit	 Copy	 Delete	P039	Energy Drink
☐	 Edit	 Copy	 Delete	P040	Mineral Water

Using GROUP BY

Query 05: Show each category with the total number of unique products and their total price.

```
1 SELECT Category,
2 COUNT(DISTINCT Product_name) AS 'Item_name',
3 SUM(Price) AS Total_Price
4 FROM product
5 GROUP BY Category;
```

Category	Item_name	Total_Price
Beverage	6	1000.00
Cleaning	3	660.00
Dairy	5	1020.00
Grocery	10	2345.00
Hygiene	3	340.00
Personal Care	5	755.00
Snacks	4	240.00
Spices	4	340.00

Using Aggregated Function

Query 06: (a) What was the total quantity of items in each cancelled cart?

```

1 SELECT Cart_ID , SUM(Quantity) AS 'Total_cancelled_items'
2 FROM cart
3 WHERE Status = 'Cancelled'
4 Group by Cart_ID;

```

←T→ ▼					Cart_ID	Total_cancelled_items
☐	 Edit	 Copy	 Delete	A004	2	
☐	 Edit	 Copy	 Delete	A008	1	
☐	 Edit	 Copy	 Delete	A013	1	
☐	 Edit	 Copy	 Delete	A017	2	
☐	 Edit	 Copy	 Delete	A023	10	
☐	 Edit	 Copy	 Delete	A025	2	

(b) Show the User's name and ID with their total cancelled orders.

```

1 |SELECT U.User_ID, U.Last_name,
2 |SUM(C.Quantity) AS 'Total_cancelled_items'
3 |FROM user AS U
4 |INNER JOIN cart AS C
5 |ON ( U.User_ID = C.User_ID )
6 |WHERE C.Status = 'Cancelled'
7 |Group by U.User_ID, U.Last_name;

```

User_ID	Last_name	Total_cancelled_items
2	Hossain	1
4	Ahmed	1
5	Jahan	2
11	Hasan	10
18	Chowdhury	2
20	Anjum	2

Using Sub-Query

Query 07: Which product/s have a price more than or equal to the Dairy product whose ID is P006?

```
1 SELECT Product_name
2 FROM product
3 Where Price >= (
4     SELECT Price
5     FROM product
6     WHERE Category = 'Dairy' AND Product_ID = 'P006' );
```

Extra options

Product_name

Rice 5kg

Milk Powder 500g

Coffee Jar

Olive Oil

THE END